

OIS-014 — Reference Model

Status: Foundational

Authority: Implementation

Depends on:

- OIS-000 through OIS-013
 - OAS-000 through OAS-031
 - OCS-000 through OCS-032
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Purpose

This specification defines the Organodynamics Reference Model.

The Reference Model is the canonical implementation abstraction from which all conforming implementations may be derived.

The Reference Model is not executable.

It is not a software architecture.

It is not a reference implementation.

It is the implementation-independent structural model that explains how constitutional responsibilities cooperate to produce executable constitutional behavior.

1. Implementation Objective

The objective of the Reference Model is to establish a common engineering vocabulary for independently developed implementations.

The Model SHALL preserve:

Constitutional responsibilities.

Architectural boundaries.

Operational relationships.

Behavioral equivalence.

Implementation independence.

2. Model Structure

The Reference Model SHALL describe:

Constitutional Layer.

Architectural Layer.

Implementation Layer.

Operational Layer.

Infrastructure Layer.

Each layer SHALL preserve explicit responsibilities.

3. Canonical Components

The Reference Model SHALL include canonical implementation components.

Typical components include:

Identity Resolution.

Schema Registry.

Event Store.

Conversation Engine.

Knowledge Engine.

Decision Engine.

Federation Engine.

Verification Services.

Recovery Services.

Future canonical components MAY be introduced while preserving existing constitutional relationships.

4. Component Relationships

Every component SHALL expose explicit operational relationships.

Relationships SHALL define:

Dependencies.

Responsibilities.

Observable interactions.

Constitutional contracts.

Relationship direction SHALL remain explicit.

5. Information Flow

The Reference Model SHALL describe constitutional information flow.

Typical flow SHALL preserve:

Identity.

Events.

Relationships.

Reasoning.

Knowledge.

Decisions.

Learning.

Implementation optimizations SHALL preserve equivalent flow.

6. Operational Boundaries

Every canonical component SHALL define:

Responsibilities.

Observable behavior.

Inputs.

Outputs.

Constitutional invariants.

Operational boundaries SHALL remain explicit.

7. Replaceability

Every canonical component SHALL be independently replaceable.

Replacement SHALL preserve:

Behavior.

Identity.

Architectural compatibility.

Constitutional equivalence.

Replaceability SHALL remain demonstrable.

8. Extensibility

The Reference Model SHALL permit additional components.

Extensions SHALL:

Declare responsibilities.

Preserve existing contracts.

Respect constitutional boundaries.

Remain independently replaceable.

The Model SHALL remain open for future evolution.

9. Verification

Verification SHALL demonstrate:

Architectural consistency.

Behavioral equivalence.

Boundary preservation.

Operational compatibility.

Independent reconstructability.

10. Reference Model Test

The Reference Model satisfies this specification when independent engineering teams derive architecturally equivalent implementations using different technologies while preserving identical constitutional responsibilities and equivalent operational behavior.

Conformance

An implementation conforms to this specification when every implementation component can be mapped to the Reference Model while preserving constitutional responsibilities, architectural relationships, operational invariants, and behavioral equivalence.

Implementation Principle

Specifications define truth.

Architectures organize truth.

Reference Models organize implementation.

Implementations realize the Model.

The purpose of the Reference Model is therefore not to dictate software,

but to provide a common engineering language through which independently developed systems can remain constitutionally identical while remaining technologically diverse.

End of Specification OIS-014